HRMS

07 工時管理 — 系統設計書

HR07 TIMESHEET Service — System Design

系統名稱	版本
人力資源暨專案管理系統	**1.0**
日期	作者
2026-03-17	**HRMS 開發團隊**

目錄

工時管理服務系統設計書

版本: 1.1 **日期:** 2025-12-26 **Domain代號:** 07 (TSH) **導入階段:** 第二階段（專案管理核心） **目標:** 提供工程師完整的系統實作規格，供PM建立工項清單

目錄

1. 服務概述
2. UI設計
3. UX流程設計
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7. Domain設計
8. 領域事件設計
9. API設計
10. 工項清單摘要

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1. 服務概述

1.1 服務定位

工時管理服務負責員工工時回報、PM審核及工時統計分析。這是專案成本核算的關鍵數據來源，需與專案服務和薪資服務緊密整合。

1.2 核心功能

- ✅ **工時回報:** 日報/週報模式
- ✅ **工時審核:** PM審核工作流程
- ✅ **工時防呆:** 超時檢查、日期驗證、重複檢查
- ✅ **工時統計:** 個人/專案/部門統計
- ✅ **工時鎖定:** 薪資結算後自動鎖定

1.3 技術架構

- **前端:** ReactJS + Redux + Ant Design
- **後端:** Spring Boot 3.1.x + MyBatis
- **資料庫:** PostgreSQL 15.x
- **事件匯流排:** Kafka

1.4 服務邊界

屬於本服務	不屬於本服務
工時回報管理	專案定義 (Project Service)
工時審核流程	成本計算細節 (需整合薪資)
工時統計分析	員工基本資料 (Organization Service)
工時鎖定機制	

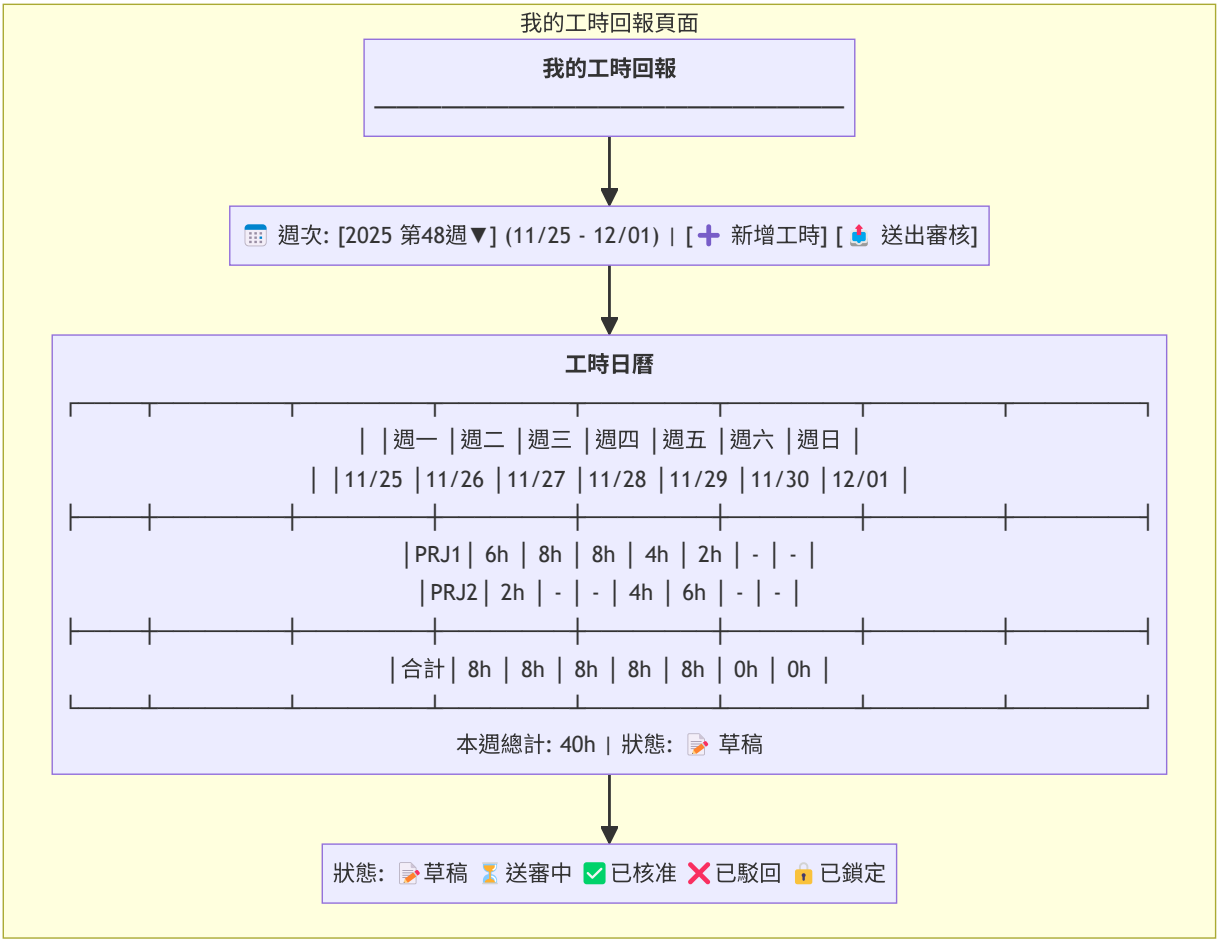
2. UI設計

2.1 頁面清單

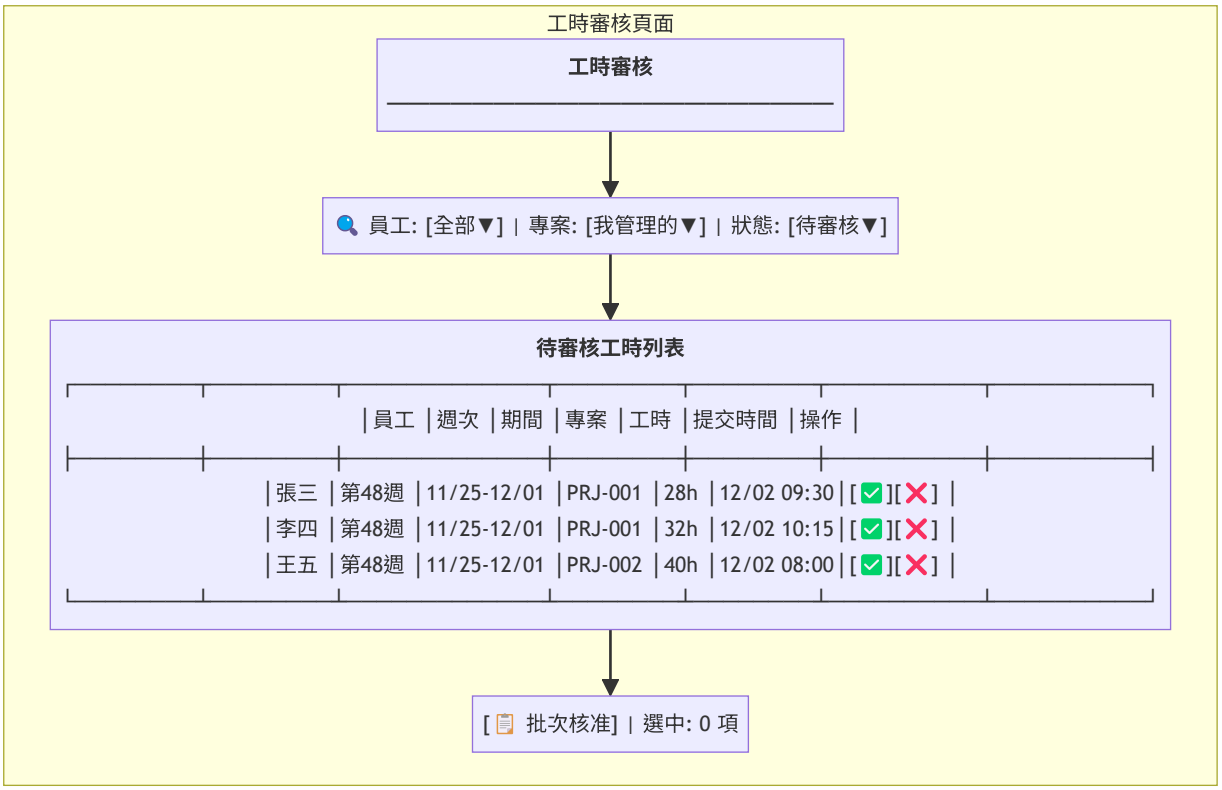
頁面代碼	頁面名稱	路由	權限要求
`HR07-P01`	我的工時回報頁面	`/profile/timesheets`	-
`HR07-P02`	工時審核頁面	`/admin/timesheets/approval`	timesheet:approve
`HR07-P03`	工時統計報表頁面	`/admin/timesheets/reports`	timesheet:report:read
`HR07-P04`	專案工時查詢頁面	`/admin/timesheets/by-project`	timesheet:read:all
`HR07-M01`	工時填報對話框	(Modal)	-
`HR07-M02`	駁回原因對話框	(Modal)	timesheet:approve

2.2 UI線稿 (Mermaid)

2.2.1 我的工時回報頁面 (HR07-P01)



2.2.2 工時審核頁面 (HR07-P02)



2.2.3 工時統計報表頁面 (HR07-P03)

```
┌─────────────────────────────────────────────────────────────────────────────┐
│                                 工時統計報表頁面                                 │
│                                 ┌──────────────────────────────┐                │
│                                 │        工時統計報表          │                │
│                                 │──────────────────────────────│                │
│                                 │                              │                │
│                                 └──────────────────────────────┘                │
│                                              │                                  │
│                                              ▼                                  │
│                 ┌──────────────────────────────────────────────────────────┐   │
│                 │ 📅 日期區間: [2025/11/01] - [2025/11/30] │ 篩選: [部門▼] [員工▼] │ [🔍 查詢] [📥 匯出] │
│                 └──────────────────────────────────────────────────────────┘   │
│                                              │                                  │
│                                              ▼                                  │
│                 ┌──────────────────────────────────────────────────────────┐   │
│                 │                        統計摘要                          │   │
│                 │ ┌──────────┬──────────┬──────────┬──────────┐            │   │
│                 │ │ 總工時   │ 專案工時 │ 平均日工時 │ 未回報人數 │            │   │
│                 │ │ 4,800h   │ 4,200h   │ 7.5h     │ 3        │            │   │
│                 │ └──────────┴──────────┴──────────┴──────────┘            │   │
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│                                              │                                  │
│                                              ▼                                  │
│                                 ┌──────────────────────────────┐                │
│                                 │ 📊 [工時趨勢圖] [專案分布] [員工排行] │                │
│                                 └──────────────────────────────┘                │
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```
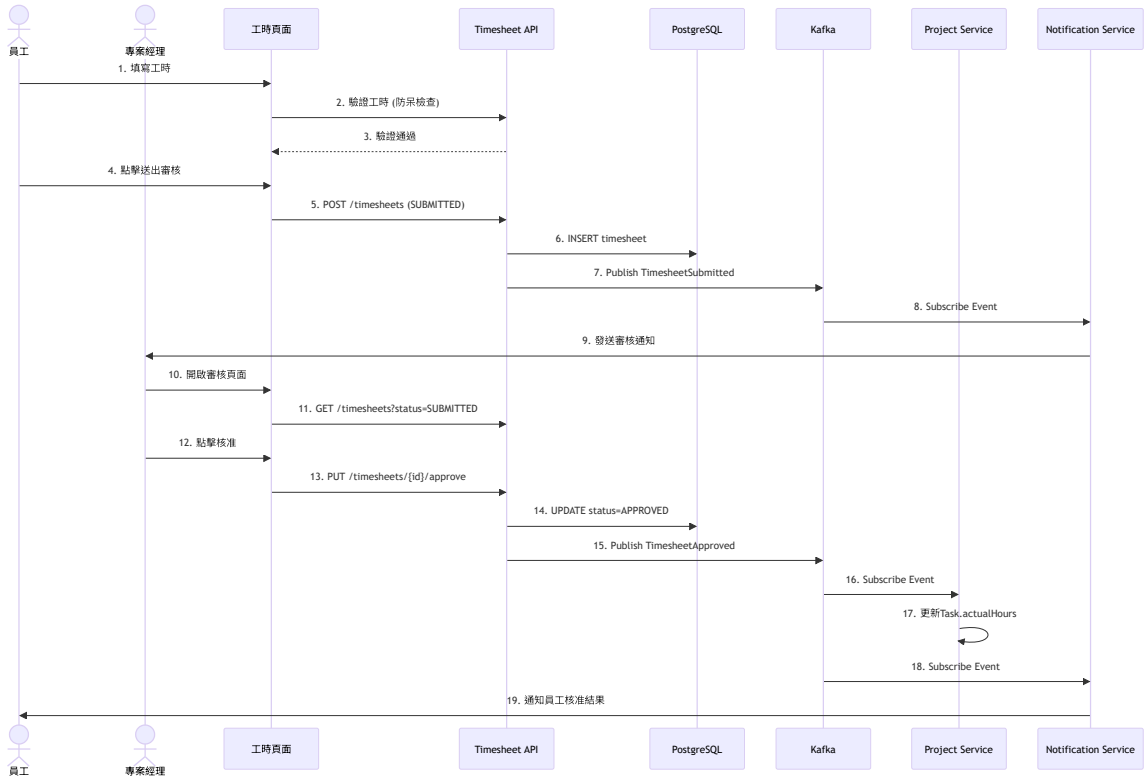
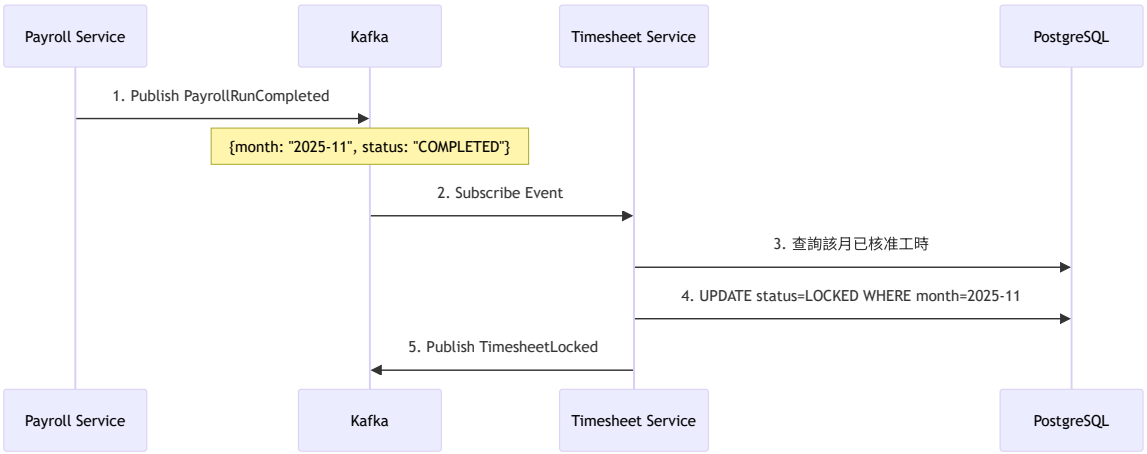

3. UX流程設計

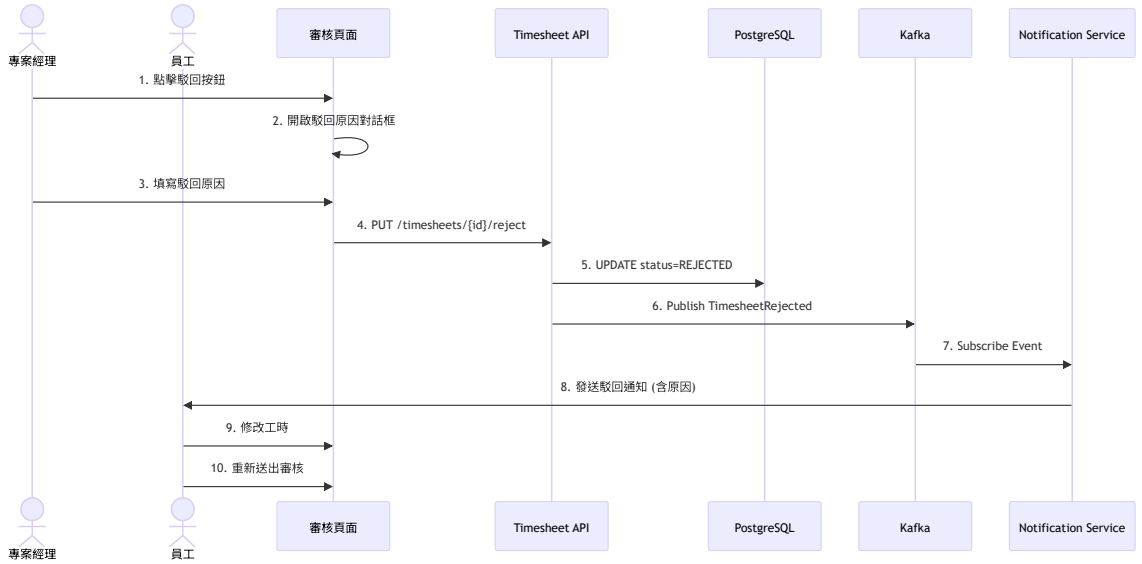
3.1 工時回報與審核流程



3.2 工時鎖定流程 (Event-Driven)



3.3 工時駁回流程


專案經理
員工
審核頁面
Timesheet API
PostgreSQL
Kafka
Notification Service
1. 點擊駁回按鈕
2. 開啟駁回原因對話框
3. 填寫駁回原因
4. PUT /timesheets/{id}/reject
5. UPDATE status=REJECTED
6. Publish TimesheetRejected
7. Subscribe Event
8. 發送駁回通知 (含原因)
9. 修改工時
10. 重新送出審核

4. 畫面事件說明

4.1 我的工時回報頁面事件 (HR07-P01)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-TSH-01`	週次選擇器	onChange	載入指定週次工時資料	GET /api/v1/timesheets/my?week={week}
`E-TSH-02`	新增工時按鈕	onClick	開啟工時填報對話框 (HR07-M01)	-
`E-TSH-03`	儲存工時按鈕	onClick	儲存工時明細	POST /api/v1/timesheets/entries
`E-TSH-04`	送出審核按鈕	onClick	提交整週工時	PUT /api/v1/timesheets/{id}/submit
`E-TSH-05`	工時格點擊	onClick	開啟編輯對話框	GET /api/v1/timesheets/entries/{id}
`E-TSH-06`	刪除工時按鈕	onClick	確認後刪除工時明細	DELETE /api/v1/timesheets/entries/{id}
`E-TSH-07`	專案下拉選單	onChange	載入可選擇的任務清單	GET /api/v1/projects/{id}/tasks

E-TSH-03 詳細流程:

```
const handleSaveEntry = async (values: TimesheetEntryFormData) => {
  try {
    // 1. 前端驗證
    if (values.hours > 24 || values.hours <= 0) {
      message.error('工時必須在0-24小時之間');
      return;
    }

    // 2. 檢查日期是否為未來日期
    if (dayjs(values.workDate).isAfter(dayjs(), 'day')) {
      message.error('不可回報未來日期的工時');
      return;
    }

    // 3. 呼叫API儲存
    await timesheetService.createEntry({
      timesheetId: currentTimesheet.id,
      projectId: values.projectId,
      taskId: values.taskId,
      workDate: values.workDate,
      hours: values.hours,
      description: values.description
    });

    // 4. 刷新列表
    await fetchTimesheetData();
    message.success('工時已儲存');
    closeModal();

  } catch (error) {
    if (error.code === 'DUPLICATE_ENTRY') {
      message.error('同日同專案已有工時記錄');
    } else if (error.code === 'EXCEED_DAILY_LIMIT') {
      message.error('單日工時不可超過24小時');
    } else {
      message.error('儲存失敗: ' + error.message);
    }
  }
};
```

4.2 工時審核頁面事件 (HR07-P02)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-TSHA-01`	員工篩選器	onChange	篩選待審核列表	GET /api/v1/timesheets/pending-approval?employeeId={id}
`E-TSHA-02`	專案篩選器	onChange	篩選待審核列表	GET /api/v1/timesheets/pending-approval?projectId={id}
`E-TSHA-03`	核准按鈕	onClick	核准單筆工時	PUT /api/v1/timesheets/{id}/approve
`E-TSHA-04`	駁回按鈕	onClick	開啟駁回原因對話框	-
`E-TSHA-05`	確認駁回	onClick	駁回工時	PUT /api/v1/timesheets/{id}/reject
`E-TSHA-06`	批次選擇核取框	onChange	更新批次選中狀態	-
`E-TSHA-07`	批次核准按鈕	onClick	批次核准工時	PUT /api/v1/timesheets/batch-approve
`E-TSHA-08`	查看詳情連結	onClick	展開工時明細	GET /api/v1/timesheets/{id}/entries

E-TSHA-03 詳細流程:

```
const handleApprove = async (timesheetId: string) => {
  try {
    // 1. 呼叫核准API
    await timesheetService.approve(timesheetId);

    // 2. 顯示成功訊息
    message.success('工時已核准');

    // 3. 刷新待審核列表
    await fetchPendingApprovals();

  } catch (error) {
    message.error('核准失敗: ' + error.message);
  }
};
```

4.3 工時統計報表頁面事件 (HR07-P03)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-TSHR-01`	日期區間選擇	onChange	更新查詢條件	-
`E-TSHR-02`	查詢按鈕	onClick	查詢統計資料	GET /api/v1/timesheets/summary
`E-TSHR-03`	匯出按鈕	onClick	匯出Excel報表	GET /api/v1/timesheets/export
`E-TSHR-04`	圖表切換Tab	onClick	切換圖表類型	-

5. Data Flow設計

5.1 前端狀態管理 (Redux)

5.1.1 State結構

```
interface TimesheetState {
  // 我的工時
  myTimesheet: {
    currentWeek: string;  // 格式: "2025-W48"
    timesheet: Timesheet | null;
    entries: TimesheetEntry[];
    loading: boolean;
    error: string | null;
  };

  // 待審核列表
  pendingApprovals: {
    list: TimesheetSummary[];
    filters: {
      employeeId: string | null;
      projectId: string | null;
      status: string;
    };
    selectedIds: string[];
    loading: boolean;
    total: number;
    page: number;
    pageSize: number;
  };

  // 統計報表
  reports: {
    summary: ReportSummary | null;
    chartData: ChartData[];
    filters: {
      startDate: string;
      endDate: string;
      departmentId: string | null;
      employeeId: string | null;
    };
    loading: boolean;
  };
}

interface Timesheet {
  timesheetId: string;
  employeeId: string;
  periodType: 'DAILY' | 'WEEKLY';
  periodStartDate: string;
```

```
  periodEndDate: string;
  totalHours: number;
  status: TimesheetStatus;
  submittedAt: string | null;
  approvedBy: string | null;
  approvedAt: string | null;
  rejectionReason: string | null;
}

interface TimesheetEntry {
  entryId: string;
  projectId: string;
  projectName: string;
  taskId: string | null;
  taskName: string | null;
  workDate: string;
  hours: number;
  description: string;
}

type TimesheetStatus = 'DRAFT' | 'SUBMITTED' | 'APPROVED' | 'REJECTED' | 'LOCKED';
```

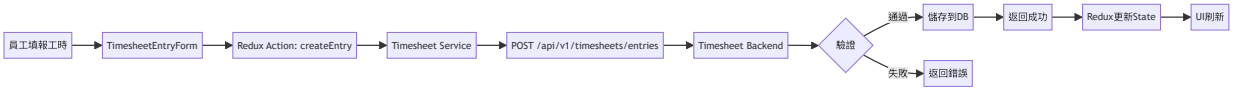
5.1.2 Redux Actions

```
// 我的工時 Actions
export const myTimesheetActions = {
  fetchMyTimesheet: createAsyncThunk(
    'timesheet/fetchMyTimesheet',
    async (week: string) => {
      const response = await timesheetService.getMyTimesheet(week);
      return response;
    }
  ),

  createEntry: createAsyncThunk(
    'timesheet/createEntry',
    async (entry: CreateEntryRequest) => {
      const response = await timesheetService.createEntry(entry);
      return response;
    }
  ),

  submitTimesheet: createAsyncThunk(
    'timesheet/submit',
    async (timesheetId: string) => {
      await timesheetService.submit(timesheetId);
      return timesheetId;
    }
  ),
};

// 審核 Actions
export const approvalActions = {
  fetchPendingApprovals: createAsyncThunk(
    'timesheet/fetchPendingApprovals',
    async (params: ApprovalQueryParams) => {
      const response = await timesheetService.getPendingApprovals(params);
      return response;
    }
  ),

  approve: createAsyncThunk(
    'timesheet/approve',
    async (timesheetId: string) => {
      await timesheetService.approve(timesheetId);
      return timesheetId;
    }
  ),

  reject: createAsyncThunk(
    'timesheet/reject',
    async ({ timesheetId, reason }: { timesheetId: string; reason: string }) => {
      await timesheetService.reject(timesheetId, reason);
```

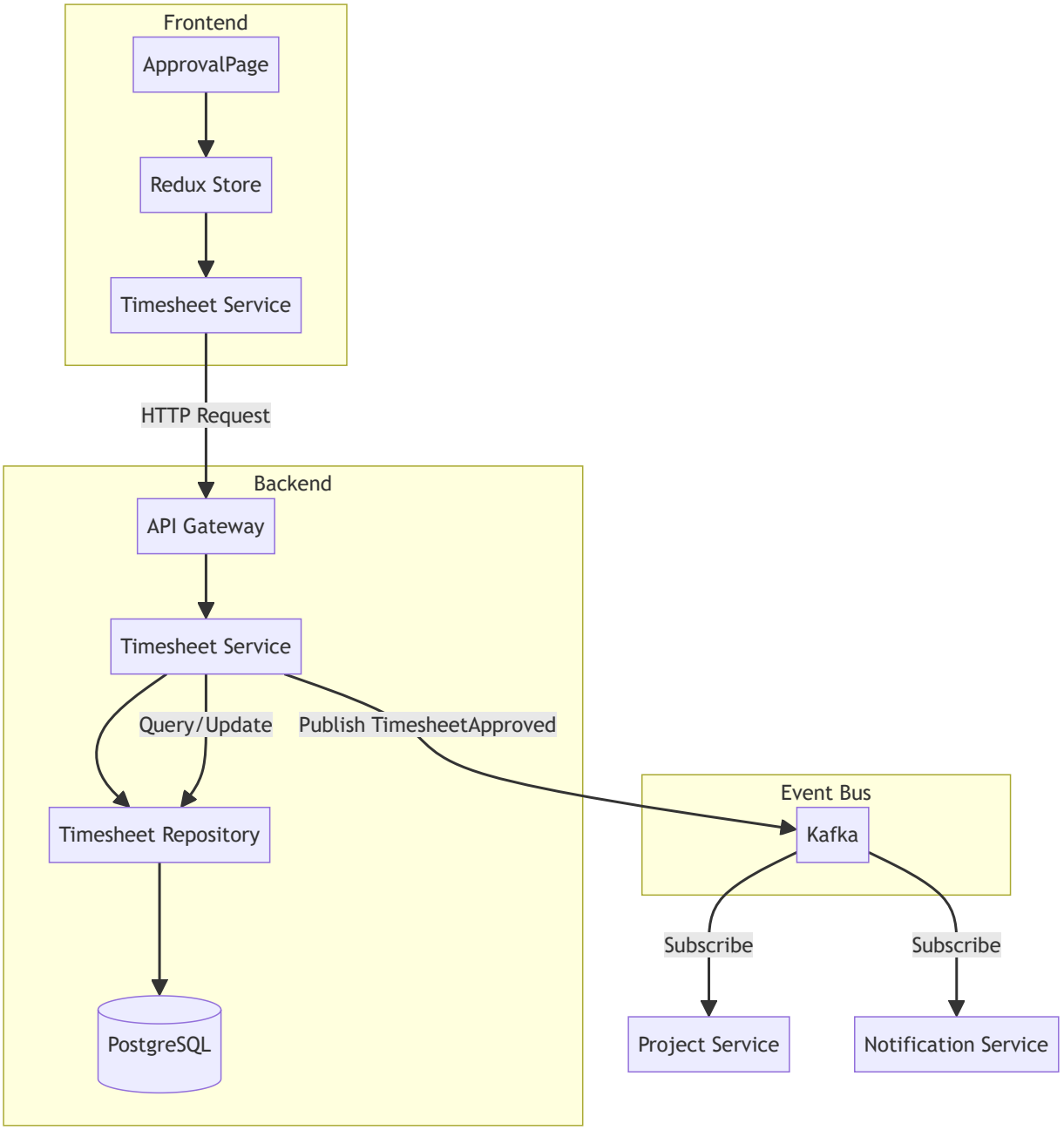
```
      return timesheetId;
    }
  ),

  batchApprove: createAsyncThunk(
    'timesheet/batchApprove',
    async (timesheetIds: string[]) => {
      await timesheetService.batchApprove(timesheetIds);
      return timesheetIds;
    }
  ),
};
```

5.2 前後端資料流

5.2.1 工時填報資料流

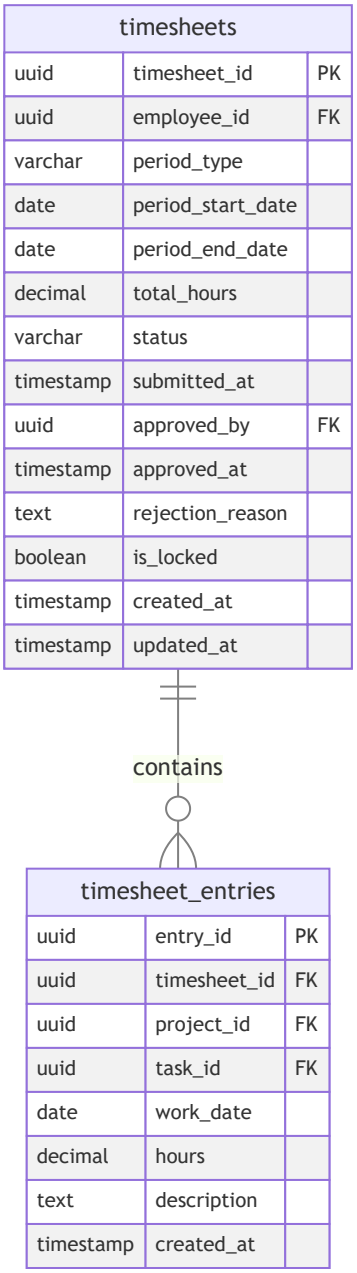


5.2.2 工時審核資料流


Frontend
ApprovalPage
Redux Store
Timesheet Service
HTTP Request
Backend
API Gateway
Timesheet Service
Query/Update
Publish TimesheetApproved
Timesheet Repository
PostgreSQL
Event Bus
Kafka
Subscribe
Subscribe
Project Service
Notification Service

6. 資料庫設計

6.1 ER Diagram



6.2 DDL Script

```sql
-- 工時表
CREATE TABLE timesheets (
    timesheet_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    employee_id UUID NOT NULL,
    period_type VARCHAR(20) NOT NULL CHECK (period_type IN ('DAILY', 'WEEKLY')),
    period_start_date DATE NOT NULL,
    period_end_date DATE NOT NULL,
    total_hours DECIMAL(6,2) DEFAULT 0,
    status VARCHAR(20) NOT NULL DEFAULT 'DRAFT'
        CHECK (status IN ('DRAFT', 'SUBMITTED', 'APPROVED', 'REJECTED', 'LOCKED')),
    submitted_at TIMESTAMP,
    approved_by UUID,
    approved_at TIMESTAMP,
    rejection_reason TEXT,
    is_locked BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT uk_timesheet_period UNIQUE (employee_id, period_start_date, period_end_date)
);

CREATE INDEX idx_timesheet_employee ON timesheets(employee_id);
CREATE INDEX idx_timesheet_status ON timesheets(status);
CREATE INDEX idx_timesheet_period ON timesheets(period_start_date, period_end_date);
CREATE INDEX idx_timesheet_approved_by ON timesheets(approved_by);

COMMENT ON TABLE timesheets IS '工時表頭';
COMMENT ON COLUMN timesheets.period_type IS '週期類型: DAILY-日報, WEEKLY-週報';
COMMENT ON COLUMN timesheets.status IS '狀態: DRAFT/SUBMITTED/APPROVED/REJECTED/LOCKED';

-- 工時明細表
CREATE TABLE timesheet_entries (
    entry_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    timesheet_id UUID NOT NULL REFERENCES timesheets(timesheet_id) ON DELETE CASCADE,
    project_id UUID NOT NULL,
    task_id UUID,
    work_date DATE NOT NULL,
    hours DECIMAL(4,2) NOT NULL CHECK (hours > 0 AND hours <= 24),
    description TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT uk_entry UNIQUE (timesheet_id, project_id, work_date)
);

CREATE INDEX idx_entry_timesheet ON timesheet_entries(timesheet_id);
CREATE INDEX idx_entry_project ON timesheet_entries(project_id);
CREATE INDEX idx_entry_task ON timesheet_entries(task_id);
CREATE INDEX idx_entry_date ON timesheet_entries(work_date);
```

```sql
COMMENT ON TABLE timesheet_entries IS '工時明細';
COMMENT ON COLUMN timesheet_entries.hours IS '工作時數 (最大24小時)';

-- 工時統計快照表 (用於報表加速)
CREATE TABLE timesheet_summaries (
    summary_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    employee_id UUID NOT NULL,
    project_id UUID NOT NULL,
    summary_month DATE NOT NULL,
    total_hours DECIMAL(8,2) NOT NULL,
    approved_hours DECIMAL(8,2) DEFAULT 0,
    pending_hours DECIMAL(8,2) DEFAULT 0,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT uk_summary UNIQUE (employee_id, project_id, summary_month)
);

CREATE INDEX idx_summary_employee ON timesheet_summaries(employee_id);
CREATE INDEX idx_summary_project ON timesheet_summaries(project_id);
CREATE INDEX idx_summary_month ON timesheet_summaries(summary_month);
```

6.3 資料字典

表名	說明	預估資料量	成長速度	保留策略
`timesheets`	工時表頭	10,000	月增800筆	永久保留
`timesheet_entries`	工時明細	100,000	月增8,000筆	永久保留
`timesheet_summaries`	工時統計快照	5,000	月增400筆	永久保留

7. Domain設計

7.1 Timesheet聚合根

```
@Entity
@Table(name = "timesheets")
public class Timesheet {
    @EmbeddedId
    private TimesheetId id;

    @Column(name = "employee_id", nullable = false)
    private UUID employeeId;

    @Enumerated(EnumType.STRING)
    @Column(name = "period_type")
    private TimesheetPeriod periodType;

    @Column(name = "period_start_date")
    private LocalDate periodStartDate;

    @Column(name = "period_end_date")
    private LocalDate periodEndDate;

    @OneToMany(cascade = CascadeType.ALL, orphanRemoval = true)
    @JoinColumn(name = "timesheet_id")
    private List<TimesheetEntry> entries = new ArrayList<>();

    @Column(name = "total_hours")
    private BigDecimal totalHours = BigDecimal.ZERO;

    @Enumerated(EnumType.STRING)
    @Column(name = "status")
    private TimesheetStatus status;

    @Column(name = "submitted_at")
    private LocalDateTime submittedAt;

    @Column(name = "approved_by")
    private UUID approvedBy;

    @Column(name = "approved_at")
    private LocalDateTime approvedAt;

    @Column(name = "rejection_reason")
    private String rejectionReason;

    @Column(name = "is_locked")
    private boolean isLocked;

    // ========== Factory Method ==========
```

```java
    public static Timesheet create(UUID employeeId, LocalDate weekStartDate) {
        Timesheet ts = new Timesheet();
        ts.id = TimesheetId.generate();
        ts.employeeId = employeeId;
        ts.periodType = TimesheetPeriod.WEEKLY;
        ts.periodStartDate = weekStartDate;
        ts.periodEndDate = weekStartDate.plusDays(6);
        ts.status = TimesheetStatus.DRAFT;
        ts.isLocked = false;
        return ts;
    }

    // ========== Domain行為 ==========

    /**
     * 新增工時明細
     */
    public void addEntry(TimesheetEntry entry) {
        // 防呆: 工時表已鎖定
        if (this.isLocked) {
            throw new DomainException("工時表已鎖定，無法修改");
        }

        // 防呆: 只有草稿或駁回狀態可以新增
        if (this.status != TimesheetStatus.DRAFT &&
            this.status != TimesheetStatus.REJECTED) {
            throw new DomainException("目前狀態不允許新增工時");
        }

        // 防呆: 不可回報未來日期
        if (entry.getWorkDate().isAfter(LocalDate.now())) {
            throw new DomainException("不可回報未來日期的工時");
        }

        // 防呆: 日期必須在週期內
        if (entry.getWorkDate().isBefore(this.periodStartDate) ||
            entry.getWorkDate().isAfter(this.periodEndDate)) {
            throw new DomainException("工作日期必須在週期範圍內");
        }

        // 防呆: 同日同專案不可重複
        boolean duplicate = this.entries.stream()
            .anyMatch(e -> e.getWorkDate().equals(entry.getWorkDate())
                && e.getProjectId().equals(entry.getProjectId()));
        if (duplicate) {
            throw new DomainException("同日同專案不可重複回報");
        }

        // 防呆: 每日不超過24小時
        BigDecimal dayTotal = this.entries.stream()
```

```java
            .filter(e -> e.getWorkDate().equals(entry.getWorkDate()))
            .map(TimesheetEntry::getHours)
            .reduce(BigDecimal.ZERO, BigDecimal::add);

        if (dayTotal.add(entry.getHours()).compareTo(new BigDecimal("24")) > 0) {
            throw new DomainException("單日工時不可超過24小時");
        }

        this.entries.add(entry);
        recalculateTotal();
    }

    /**
     * 刪除工時明細
     */
    public void removeEntry(UUID entryId) {
        if (this.isLocked) {
            throw new DomainException("工時表已鎖定，無法修改");
        }

        if (this.status != TimesheetStatus.DRAFT &&
            this.status != TimesheetStatus.REJECTED) {
            throw new DomainException("目前狀態不允許刪除工時");
        }

        this.entries.removeIf(e -> e.getId().equals(entryId));
        recalculateTotal();
    }

    /**
     * 提交審核
     */
    public void submit() {
        if (this.status != TimesheetStatus.DRAFT &&
            this.status != TimesheetStatus.REJECTED) {
            throw new DomainException("只有草稿或已駁回狀態可以提交");
        }

        if (this.entries.isEmpty()) {
            throw new DomainException("至少需要一筆工時記錄");
        }

        this.status = TimesheetStatus.SUBMITTED;
        this.submittedAt = LocalDateTime.now();
        this.rejectionReason = null; // 清除之前的駁回原因

        DomainEventPublisher.publish(new TimesheetSubmittedEvent(
            this.id.getValue(),
            this.employeeId,
            this.totalHours,
            this.periodStartDate,
```

```
            this.periodEndDate
        ));
    }

    /**
     * 核准
     */
    public void approve(UUID approverId) {
        if (this.status != TimesheetStatus.SUBMITTED) {
            throw new DomainException("只有已提交狀態可以核准");
        }

        this.status = TimesheetStatus.APPROVED;
        this.approvedBy = approverId;
        this.approvedAt = LocalDateTime.now();

        // 彙整各專案工時
        Map<UUID, BigDecimal> projectHours = this.entries.stream()
            .collect(Collectors.groupingBy(
                TimesheetEntry::getProjectId,
                Collectors.reducing(BigDecimal.ZERO, TimesheetEntry::getHours, BigDecimal::
            ));

        DomainEventPublisher.publish(new TimesheetApprovedEvent(
            this.id.getValue(),
            this.employeeId,
            projectHours
        ));
    }

    /**
     * 駁回
     */
    public void reject(UUID approverId, String reason) {
        if (this.status != TimesheetStatus.SUBMITTED) {
            throw new DomainException("只有已提交狀態可以駁回");
        }

        Objects.requireNonNull(reason, "駁回原因不可為空");

        this.status = TimesheetStatus.REJECTED;
        this.approvedBy = approverId;
        this.approvedAt = LocalDateTime.now();
        this.rejectionReason = reason;

        DomainEventPublisher.publish(new TimesheetRejectedEvent(
            this.id.getValue(),
            this.employeeId,
            reason
        ));
    }
```

```java
    /**
     * 鎖定（薪資結算後）
     */
    public void lock() {
        if (this.status != TimesheetStatus.APPROVED) {
            throw new DomainException("只有已核准狀態可以鎖定");
        }
        this.status = TimesheetStatus.LOCKED;
        this.isLocked = true;

        DomainEventPublisher.publish(new TimesheetLockedEvent(
            this.id.getValue(),
            this.employeeId
        ));
    }

    private void recalculateTotal() {
        this.totalHours = this.entries.stream()
            .map(TimesheetEntry::getHours)
            .reduce(BigDecimal.ZERO, BigDecimal::add);
    }
}
```

7.2 TimesheetEntry實體

```java
@Entity
@Table(name = "timesheet_entries")
public class TimesheetEntry {
    @EmbeddedId
    private EntryId id;

    @Column(name = "project_id", nullable = false)
    private UUID projectId;

    @Column(name = "task_id")
    private UUID taskId;

    @Column(name = "work_date", nullable = false)
    private LocalDate workDate;

    @Column(name = "hours", nullable = false)
    private BigDecimal hours;

    @Column(name = "description")
    private String description;

    public static TimesheetEntry create(UUID projectId, UUID taskId,
            LocalDate workDate, BigDecimal hours, String description) {

        if (hours.compareTo(BigDecimal.ZERO) <= 0) {
            throw new DomainException("工時必須大於0");
        }

        if (hours.compareTo(new BigDecimal("24")) > 0) {
            throw new DomainException("單筆工時不可超過24小時");
        }

        TimesheetEntry entry = new TimesheetEntry();
        entry.id = EntryId.generate();
        entry.projectId = projectId;
        entry.taskId = taskId;
        entry.workDate = workDate;
        entry.hours = hours;
        entry.description = description;
        return entry;
    }
}
```

7.3 值對象

```java
public enum TimesheetStatus {
    DRAFT("草稿", true),
    SUBMITTED("已提交", false),
    APPROVED("已核准", false),
    REJECTED("已駁回", true),
    LOCKED("已鎖定", false);

    private final String displayName;
    private final boolean editable;

    TimesheetStatus(String displayName, boolean editable) {
        this.displayName = displayName;
        this.editable = editable;
    }

    public boolean isEditable() {
        return editable;
    }
}

public enum TimesheetPeriod {
    DAILY("日報"),
    WEEKLY("週報");

    private final String displayName;

    TimesheetPeriod(String displayName) {
        this.displayName = displayName;
    }
}
```

8. 領域事件設計

8.1 事件清單

事件名稱	觸發時機	發布服務	訂閱服務
`TimesheetCreatedEvent`	建立工時表	Timesheet	-
`TimesheetSubmittedEvent`	提交工時	Timesheet	Workflow, Notification
`TimesheetApprovedEvent`	核准工時	Timesheet	Project, Payroll, Notification
`TimesheetRejectedEvent`	駁回工時	Timesheet	Notification
`TimesheetLockedEvent`	鎖定工時	Timesheet	-

8.2 事件Schema

TimesheetSubmittedEvent

```json
{
  "eventId": "uuid",
  "eventType": "TimesheetSubmittedEvent",
  "occurredAt": "2025-12-02T09:30:00Z",
  "aggregateId": "timesheet-uuid",
  "aggregateType": "Timesheet",
  "payload": {
    "timesheetId": "uuid",
    "employeeId": "uuid",
    "totalHours": 40.0,
    "periodStartDate": "2025-11-25",
    "periodEndDate": "2025-12-01"
  }
}
```

TimesheetApprovedEvent

```
{
  "eventId": "uuid",
  "eventType": "TimesheetApprovedEvent",
  "occurredAt": "2025-12-03T14:00:00Z",
  "aggregateId": "timesheet-uuid",
  "aggregateType": "Timesheet",
  "payload": {
    "timesheetId": "uuid",
    "employeeId": "uuid",
    "approvedBy": "manager-uuid",
    "projectHours": {
      "project-uuid-1": 28.0,
      "project-uuid-2": 12.0
    }
  }
}
```

TimesheetRejectedEvent

```
{
  "eventId": "uuid",
  "eventType": "TimesheetRejectedEvent",
  "occurredAt": "2025-12-03T14:00:00Z",
  "aggregateId": "timesheet-uuid",
  "aggregateType": "Timesheet",
  "payload": {
    "timesheetId": "uuid",
    "employeeId": "uuid",
    "rejectedBy": "manager-uuid",
    "reason": "工時與差勤記錄不符，請確認"
  }
}
```

---

9. API設計

9.1 Controller命名對照

Controller	說明
`HR07TimesheetCmdController`	工時Command操作
`HR07TimesheetQryController`	工時Query操作
`HR07ApprovalCmdController`	審核Command操作
`HR07ApprovalQryController`	審核Query操作
`HR07ReportQryController`	報表Query操作

9.2 API總覽 (14個端點)

端點	方法	Controller	說明
`/api/v1/timesheets`	POST	HR07TimesheetCmdController	建立工時表
`/api/v1/timesheets/{id}/entries`	POST	HR07TimesheetCmdController	新增工時明細
`/api/v1/timesheets/{id}/entries/{entryId}`	PUT	HR07TimesheetCmdController	更新工時明細
`/api/v1/timesheets/{id}/entries/{entryId}`	DELETE	HR07TimesheetCmdController	刪除工時明細
`/api/v1/timesheets/{id}/submit`	PUT	HR07TimesheetCmdController	提交審核
`/api/v1/timesheets/{id}/approve`	PUT	HR07ApprovalCmdController	核准
`/api/v1/timesheets/{id}/reject`	PUT	HR07ApprovalCmdController	駁回
`/api/v1/timesheets/batch-approve`	PUT	HR07ApprovalCmdController	批次核准
`/api/v1/timesheets/my`	GET	HR07TimesheetQryController	我的工時
`/api/v1/timesheets/{id}/entries`	GET	HR07TimesheetQryController	工時明細
`/api/v1/timesheets/pending-approval`	GET	HR07ApprovalQryController	待審核列表
`/api/v1/timesheets/summary`	GET	HR07ReportQryController	個人統計
`/api/v1/timesheets/project-summary`	GET	HR07ReportQryController	專案統計
`/api/v1/timesheets/unreported`	GET	HR07ReportQryController	未回報員工

9.3 API詳細規格

9.3.1 新增工時明細

請求:

```
POST /api/v1/timesheets/{timesheetId}/entries
Content-Type: application/json
Authorization: Bearer {accessToken}

{
  "projectId": "project-uuid",
  "taskId": "task-uuid",
  "workDate": "2025-11-25",
  "hours": 8.0,
  "description": "完成需求分析文件"
}
```

成功回應:

```
{
  "code": "SUCCESS",
  "message": "工時已新增",
  "data": {
    "entryId": "entry-uuid"
  }
}
```

9.3.2 提交審核

請求:

```
PUT /api/v1/timesheets/{timesheetId}/submit
Authorization: Bearer {accessToken}
```

成功回應:

```
{
  "code": "SUCCESS",
  "message": "工時已提交審核"
}
```

9.3.3 駁回工時

請求:

```
PUT /api/v1/timesheets/{timesheetId}/reject
Content-Type: application/json
Authorization: Bearer {accessToken}

{
  "reason": "工時與差勤記錄不符，請確認11/26是否有請假"
}
```

成功回應:

```
{
  "code": "SUCCESS",
  "message": "工時已駁回"
}
```

---

10. 工項清單摘要

前端開發工項

工項編號	工項名稱	估計工時	優先順序
FE-07-01	HR07-P01 我的工時回報頁面 (週曆視圖)	24h	P0
FE-07-02	HR07-P02 工時審核頁面	16h	P0
FE-07-03	HR07-P03 工時統計報表頁面	16h	P1
FE-07-04	HR07-P04 專案工時查詢頁面	8h	P1
FE-07-05	HR07-M01 工時填報對話框	8h	P0
FE-07-06	HR07-M02 駁回原因對話框	4h	P0
FE-07-07	Redux Timesheet模組	8h	P0
FE-07-08	TimesheetViewModelFactory	4h	P0
FE-07-09	工時日曆元件	16h	P0
小計		**104h**	

後端開發工項

工項編號	工項名稱	估計工時	優先順序
BE-07-01	Timesheet聚合根與Repository	16h	P0
BE-07-02	TimesheetEntry實體	8h	P0
BE-07-03	工時防呆驗證Service	8h	P0
BE-07-04	工時統計Service	8h	P1
BE-07-05	工時填報API (5端點)	16h	P0
BE-07-06	工時審核API (4端點)	12h	P0
BE-07-07	工時報表API (3端點)	8h	P1
BE-07-08	訂閱PayrollRunCompleted事件鎖定工時	4h	P0
BE-07-09	領域事件發布	4h	P0
BE-07-10	Swagger文件	2h	P1
小計		**86h**	

資料庫開發工項

工項編號	工項名稱	估計工時	優先順序
DB-07-01	建立3個資料表DDL	2h	P0
DB-07-02	建立索引	1h	P0
DB-07-03	初始化測試資料	1h	P1
小計		**4h**	

測試開發工項

工項編號	工項名稱	估計工時	優先順序
TE-07-01	Timesheet聚合根單元測試	8h	P0
TE-07-02	工時防呆邏輯測試	8h	P0
TE-07-03	工時API整合測試	8h	P0
TE-07-04	前端Factory測試	4h	P0
TE-07-05	前端Component測試	8h	P1
小計		**36h**	

總計

類別	工時
前端開發	104h
後端開發	86h
資料庫開發	4h
測試開發	36h
合計	**230h**

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